

1.2 The degree of a vertex

Let $G = (V, E)$ be a (non-empty) graph. The set of neighbours of a vertex v in G is denoted by $N_G(v)$, or briefly by $N(v)$.¹ More generally for $U \subseteq V$, the neighbours in $V \setminus U$ of vertices in U are called *neighbours of U* ; their set is denoted by $N(U)$. $N(v)$

The *degree* (or *valency*) $d_G(v) = d(v)$ of a vertex v is the number *degree $d(v)$*
 $|E(v)|$ of edges at v ; by our definition of a graph,² this is equal to the
 number of neighbours of v . A vertex of degree 0 is *isolated*. The number *isolated*
 $\delta(G) := \min \{ d(v) \mid v \in V \}$ is the *minimum degree* of G , the number $\delta(G)$
 $\Delta(G) := \max \{ d(v) \mid v \in V \}$ its *maximum degree*. If all the vertices $\Delta(G)$
 of G have the same degree k , then G is *k -regular*, or simply *regular*. A *regular*
 3-regular graph is called *cubic*. *cubic*

The number

$$d(G) := \frac{1}{|V|} \sum_{v \in V} d(v) \quad d(G)$$

is the *average degree* of G . Clearly, *average degree*

$$\delta(G) \leq d(G) \leq \Delta(G).$$

The average degree quantifies globally what is measured locally by the vertex degrees: the number of edges of G per vertex. Sometimes it will be convenient to express this ratio directly, as $\varepsilon(G) := |E|/|V|$. $\varepsilon(G)$

The quantities d and ε are, of course, intimately related. Indeed, if we sum up all the vertex degrees in G , we count every edge exactly twice: once from each of its ends. Thus

$$|E| = \frac{1}{2} \sum_{v \in V} d(v) = \frac{1}{2} d(G) \cdot |V|,$$

and therefore

$$\varepsilon(G) = \frac{1}{2} d(G).$$

Proposition 1.2.1. *The number of vertices of odd degree in a graph is always even.* [10.3.1]

Proof. As $|E| = \frac{1}{2} \sum_{v \in V} d(v)$ is an integer, $\sum_{v \in V} d(v)$ is even. $\square$

¹ Here, as elsewhere, we drop the index referring to the underlying graph if the reference is clear.

² but not for multigraphs; see Section 1.10

If a graph has large minimum degree, i.e. everywhere, locally, many edges per vertex, it also has many edges per vertex globally: $\varepsilon(G) = \frac{1}{2}d(G) \geq \frac{1}{2}\delta(G)$. Conversely, of course, its average degree may be large even when its minimum degree is small. However, the vertices of large degree cannot be scattered completely among vertices of small degree: as the next proposition shows, every graph G has a subgraph whose average degree is no less than the average degree of G , and whose minimum degree is more than half its average degree:

[1.4.3]
[7.2.2]

Proposition 1.2.2. *Every graph G with at least one edge has a subgraph H with $\delta(H) > \varepsilon(H) \geq \varepsilon(G)$.*

Proof. To construct H from G , let us try to delete vertices of small degree one by one, until only vertices of large degree remain. Up to which degree $d(v)$ can we afford to delete a vertex v , without lowering ε ? Clearly, up to $d(v) = \varepsilon$: then the number of vertices decreases by 1 and the number of edges by at most ε , so the overall ratio ε of edges to vertices will not decrease.

Formally, we construct a sequence $G = G_0 \supseteq G_1 \supseteq \dots$ of induced subgraphs of G as follows. If G_i has a vertex v_i of degree $d(v_i) \leq \varepsilon(G_i)$, we let $G_{i+1} := G_i - v_i$; if not, we terminate our sequence and set $H := G_i$. By the choices of v_i we have $\varepsilon(G_{i+1}) \geq \varepsilon(G_i)$ for all i , and hence $\varepsilon(H) \geq \varepsilon(G)$.

What else can we say about the graph H ? Since $\varepsilon(K^1) = 0 < \varepsilon(G)$, none of the graphs in our sequence is trivial, so in particular $H \neq \emptyset$. The fact that H has no vertex suitable for deletion thus implies $\delta(H) > \varepsilon(H)$, as claimed. $\square$

1.3 Paths and cycles

path A *path* is a non-empty graph $P = (V, E)$ of the form

$$V = \{x_0, x_1, \dots, x_k\} \quad E = \{x_0x_1, x_1x_2, \dots, x_{k-1}x_k\},$$

length P^k where the x_i are all distinct. The vertices x_0 and x_k are *linked* by P and are called its *endvertices* or *ends*; the vertices $x_1, \dots, x_{k-1}$ are the *inner* vertices of P . The number of edges of a path is its *length*, and the path of length k is denoted by P^k . Note that k is allowed to be zero; thus, $P^0 = K^1$.

We often refer to a path by the natural sequence of its vertices,³

³ More precisely, by one of the two natural sequences: $x_0 \dots x_k$ and $x_k \dots x_0$ denote the same path. Still, it often helps to fix one of these two orderings of $V(P)$ notationally: we may then speak of things like the ‘first’ vertex on P with a certain property, etc.